$d_1^+ \sim \frac{\partial}{\partial x}$ No 6 ~~30.06~~

30.10.2024

$d_e \sim I$ $\frac{1}{2h_1}$ $\frac{1}{4h_1}$ **[1]**

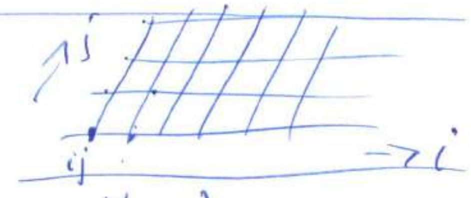
$$\begin{pmatrix} x^1_{ij} \\ x^2_{ij} \end{pmatrix} = \begin{pmatrix} x^1_0 \\ x^2_0 \end{pmatrix} + B_h \begin{pmatrix} i \\ j \end{pmatrix}$$

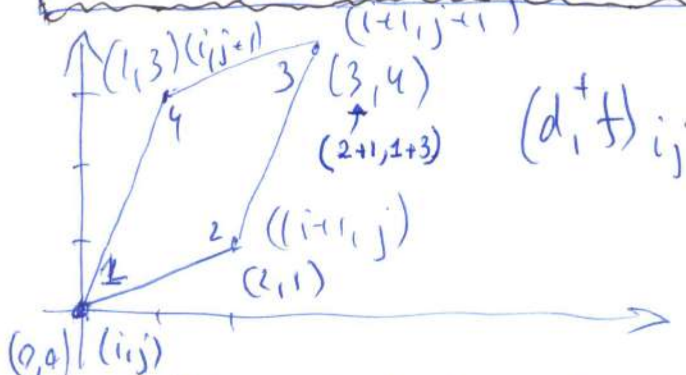
$$\begin{array}{ccc} & (2,1) & (1,3) \\ B_h = & \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} & \end{array}$$

— 4-угольная сетка

$d_0^+, \underline{d_1^+, d_2^+}, d_0^-, d_1^-, d_2^-, D_{ij} = d_i^+ d_j^-$

$$\boxed{(d_0^+ f)_{ij} = \frac{1}{4}(f_{ij} + f_{i+1\,j} + f_{i+1\,j+1} + f_{i\,j+1})}$$





$$(d_1^+ f)_{ij} = \frac{\sum_{i=1}^{4} f_i (y_{i+1} - y_{i-1})}{\sum_{i=1}^{4} x_i (y_{i+1} - y_{i-1})} =$$

$$= \frac{f_1(y_2 - y_4) + f_2(y_3 - y_1) + f_3(y_4 - y_2) + f_4(y_1 - y_3)}{x_1(y_2 - y_4) + x_2(y_3 - y_1) + x_3(y_4 - y_2) + x_4(y_1 - y_3)} =$$

$$= \frac{f_1(1-3) + f_2(4-0) + f_3(3-1) + f_4(0-4)}{0(1-3) + 2(4-0) + 3(3-1) + 1(0-4)} = \frac{-2f_1 + 4f_2 + 2f_3 - 4f_4}{10}$$

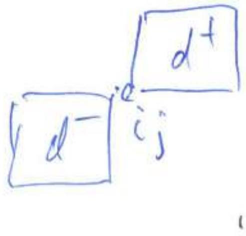
$$= \frac{1}{5}(-f_{ij} + 2f_{i+1\,j} + f_{i+1\,j+1} - 2f_{i\,j+1}) \quad \boxed{(d_1^+ f)_{ij} = \frac{1}{5}(-f_{ij} + 2f_{i+1\,j} + f_{i+1\,j+1} - 2f_{i\,j+1})}$$

$$(d_2^+ f)_{ij} = \frac{\sum_{i=1}^{4} f_i (x_{i+1} - x_{i-1})}{\sum_{i=1}^{4} y_i (x_{i+1} - x_{i-1})} = \frac{f_1(2-1) + f_2(3-0) + f_3(1-2) + f_4(0-3)}{0(2-1) + 1(3-0) + 4(1-2) + 3(0-3)}$$

3-0 0-3

$$= \frac{1}{-10}(f_1 + 3f_2 - f_3 - 3f_4)$$

$$\boxed{(d_2^+ f)_{ij} = \frac{1}{10}(-f_{ij} - 3f_{i+1\,j} + f_{i+1\,j+1} + 3f_{i\,j+1})}$$



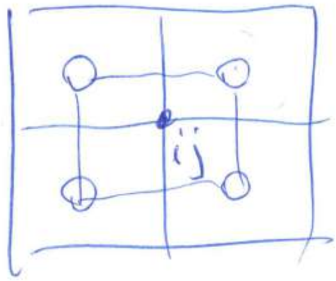
f: сопряж. опе...

$(d_0^- f)_{ij} = \frac{1}{4}(f_{ij} + f_{i-1} + f_{i-1\,j-1} + f_{i\,j-1})$

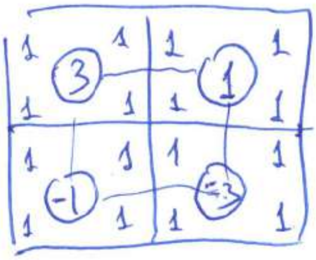
$(d_1^- f)_{ij} = \frac{1}{5}(f_{ij} - 2f_{i-1\,j} - f_{i-1\,j-1} + 2f_{i\,j-1})$

$(d_2^- f)_{ij} = \frac{1}{10}(f_{ij} + 3f_{i-1\,j} - f_{i-1\,j-1} - 3f_{i\,j-1})$

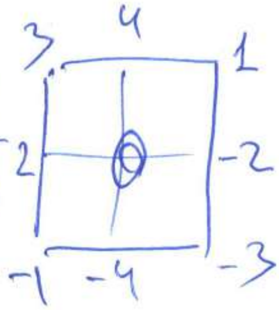
$D_{ij} = d_i^+ d_j^-$



$D_{02} = d_0^+ d_2^- = \frac{1}{4} \cdot \frac{1}{10}$

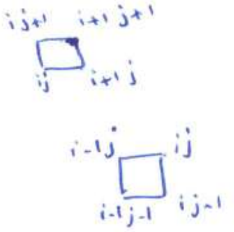


$= \frac{1}{40}$

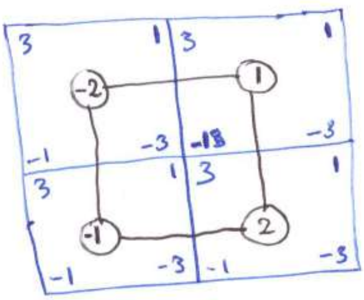


$(D_{02} f)_{ij} = \frac{1}{40}\big(3f_{i-1\,j+1} + 4f_{i\,j+1} + f_{i+1\,j+1} + 2f_{i-1\,j} - 2f_{i+1\,j} -$

$- f_{i-1\,j-1} - 4f_{i\,j-1} - 3f_{i-1\,j+1}\big)$



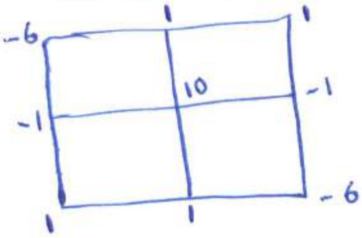
$(D_{21} f)_{ij} = d_2^+ d_1^- = \frac{1}{10} \cdot \frac{1}{5}$



$\frac{\cancel{\ }}{50}$

$-1+6+6-1$

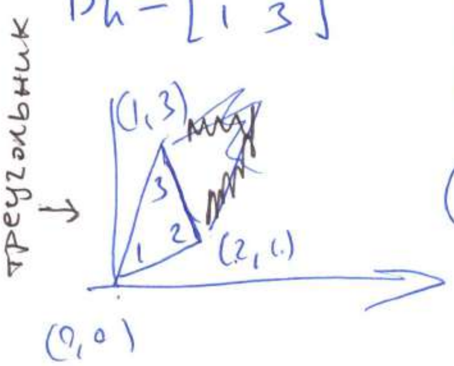
$= \frac{1}{50}$



$(D_{21} f)_{ij} = \frac{1}{50}\big[-6f_{i-1\,j+1} + f_{i\,j+1} + f_{i+1\,j+1} - f_{i-1\,j} + 10f_{ij} - f_{i+1\,j} + f_{i-1\,j-1} + f_{i\,j-1} - 6f_{i+1\,j-}$

$\boxed{3}$

$B_h = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$



$$\boxed{(d_0^+ f)_{ij} = \frac{1}{3}\left(f_{ij} + f_{i+1j} + f_{ij+1}\right)}$$

$$(d_1^+ f)_{ij} = \frac{\sum_1^3 f_i (y_{i+1} - y_{i-1})}{\sum_1^3 x_i (y_{i+1} - y_{i-1})} =$$

$$= \frac{f_1(1-3) + f_2(3-0) + f_3(0-1)}{0(1-3) + 2(3-0) + 1(0-1)} = \frac{1}{5}\left(-2f_1 + 3f_2 - f_3\right)$$

$$\boxed{(d_1^+ f)_{ij} = \frac{1}{5}\left(-2f_{ij} + 3f_{i+1j} - f_{ij+1}\right)}$$

$$(d_2^+ f)_{ij} = \frac{\sum_1^3 f_i (x_{i+1} - x_{i-1})}{\sum_1^3 y_i (x_{i+1} - x_{i-1})} = \frac{1}{-5}\left(f_1(2-1) + f_2(1-0) + f_3(0-2)\right) =$$

$$= \frac{1}{5}\left(-f_1 - f_2 + 2f_3\right)$$

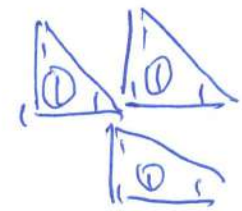
$$\boxed{(d_2^+ f)_{ij} = \frac{1}{5}\left(-f_{ij} - f_{i+1j} + 2f_{ij+1}\right)}$$

$$(d_0^- f)_{ij} = \frac{1}{3}\left(f_{ij} + f_{i-1j} + f_{ij-1}\right)$$

$$(d_1^- f)_{ij} = \frac{1}{5}\left(2f_{ij} - 3f_{i-1j} + f_{ij-1}\right)$$

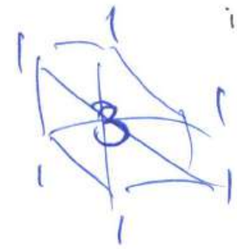
$$(d_2^- f)_{ij} = \frac{1}{5}\left(f_{ij} + f_{i-1j} - 2f_{ij-1}\right)$$

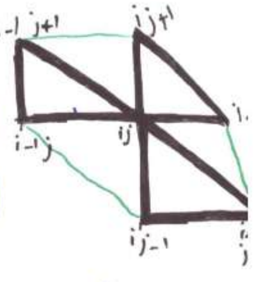
$D_{00} = \frac{1}{3}\cdot\frac{1}{3}$



$=$

$= \frac{1}{9}$

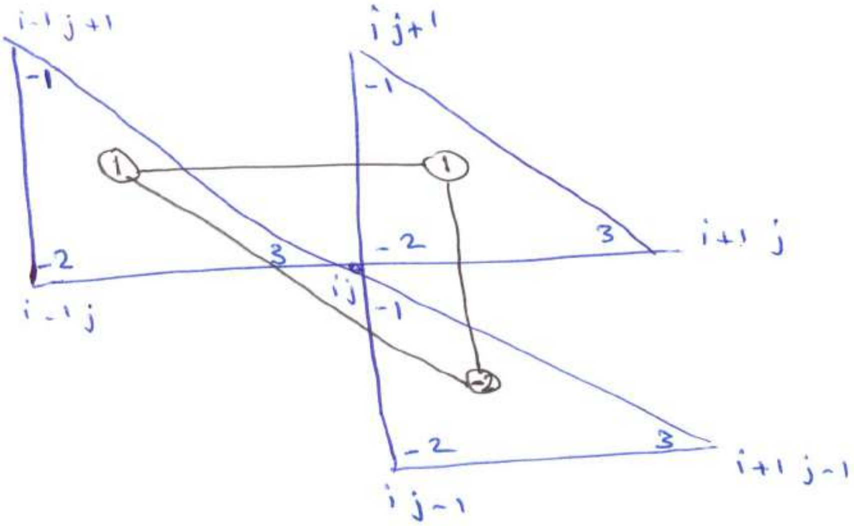


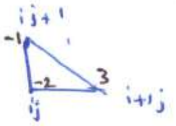


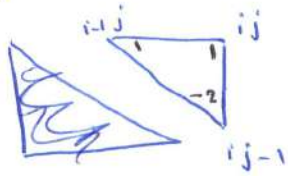
$$(D_{00} f)_{ij} = \frac{1}{9}\left(f_{i-1j+1} + f_{ij+1} + f_{i-1j} + 3f_{ij} + f_{i+1j} + f_{ij-1} + f_{i+1j-1}\right)$$

D_{12}, D_{21}

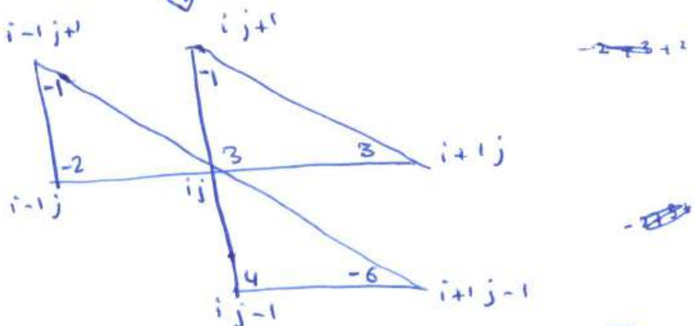
$D_{12} = d_1^+ d_2^- = \frac{1}{5} \cdot \frac{1}{5}$



$d_1^+ = \frac{1}{5}$ 

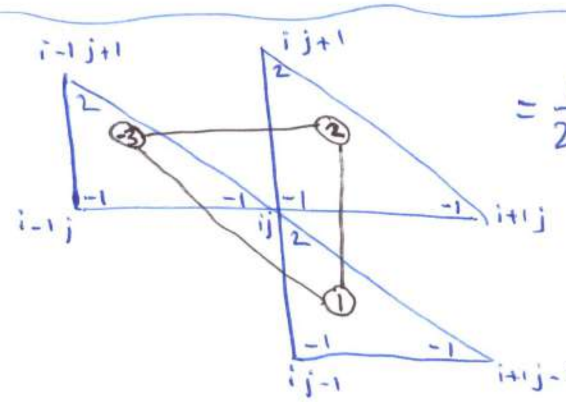
$d_2^- = \frac{1}{5}$ 

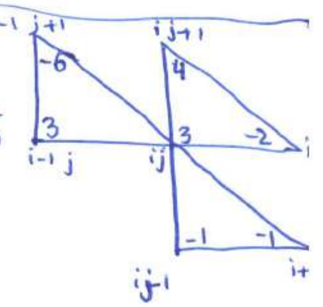
$\Downarrow$

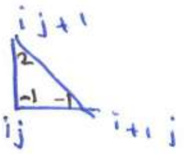


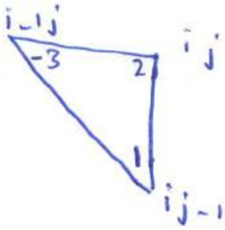
$$D_{12} = \frac{1}{25}\left[-f_{i-1\,j+1} - f_{i\,j+1} - 2f_{i-1\,j} + 3f_{ij} + 3f_{i+1\,j} + 4f_{i\,j-1} - 6f_{i+1\,j-1}\right]$$

$D_{21} = d_2^+ d_1^- = \frac{1}{5} \cdot \frac{1}{5}$



$= \frac{1}{25}$ 

$d_2^+ = \frac{1}{5}$ 

$d_1^- = \frac{1}{5}$ 

$$D_{21} = \frac{1}{25}\left[-6f_{i-1\,j+1} + 4f_{i\,j+1} + 3f_{i-1\,j} + 3f_{ij} - 2f_{i+1\,j} - f_{i\,j-1} - f_{i+1\,j-1}\right]$$

четырехугольник

ε.ъ

$\cancel{D_{00}}$

$d_0^+, d_1^+, d_2^+, d_0^-, d_1^-, d_2^-, D_{ij}$ □ $D_{12}=D_{21}$

проверить

$D_{00}, D_{11}, D_{22}, D_{01}, D_{02}, D_{12}, D_{21}$

Сит

△ $\rightarrow D_{00}, D_{11}, D_{22}, D_{01}, D_{10}, D_{02}\ D_{20}\ D_{12}\ D_{21}$

треугольник

$B_1=\begin{bmatrix}1 & -1\\ 2 & 1\end{bmatrix}$ 4x □ чет-хуг.

$B_2=\begin{bmatrix}4 & 1\\ 1 & 2\end{bmatrix}$ 3x △ треугольник

$B_1=\begin{bmatrix}4 & 1\\ 1 & 2\end{bmatrix}$ □ 4x

$B_2=\begin{bmatrix}1 & -1\\ 2 & 1\end{bmatrix}$ △ 3x

$B_1=\begin{bmatrix}2 & 1\\ 1 & 3\end{bmatrix}$ □ 4x

$B_2=\begin{bmatrix}1 & -1\\ 1 & 2\end{bmatrix}$ △ 3x

$B_1=\begin{bmatrix}1 & -1\\ 1 & 2\end{bmatrix}$ □ 4x

$B_2=\begin{bmatrix}3 & 1\\ 2 & 3\end{bmatrix}$ △ 3x

(*)

$B_1=\begin{bmatrix}2 & -1\\ 3 & 3\end{bmatrix}$ □ 4x

$B_2=\begin{bmatrix}3 & 1\\ -1 & 0\end{bmatrix}$ △ 3x

$B_1=\begin{bmatrix}2 & 0\\ -1 & 2\end{bmatrix}$ □ 4x

$B_2=\begin{bmatrix}4 & -2\\ 3 & 2\end{bmatrix}$ △

$B_1=\begin{bmatrix}3 & 1\\ -2 & 1\end{bmatrix}$ □ 4x

$B_2=\begin{bmatrix}4 & 1\\ 0 & 3\end{bmatrix}$ △ 3x

моя задача

$B_1=\begin{bmatrix}3 & 2\\ -1 & 2\end{bmatrix}$ □ 4x

$B_2=\begin{bmatrix}4 & 2\\ -2 & 2\end{bmatrix}$ △ 3x

□ : четырехугольник

△ : треугольник